

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (22 marks – approximately 40 minutes)

Giselle Beauty Limited ("GBL") specialises in production and supply of watery foaming gel for facial cleansing. It uses the first-in, first-out method of process costing. The following information relates to one of its production processes for last month:

- (1) Opening and closing work in process ("WIP") of last month were 50,000 units and 20,000 units, respectively, and their breakdowns were:

	Opening WIP		Closing WIP
	Degree of Completion	HK\$	Degree of Completion
Materials transferred-in from previous process	100%	100,000	100%
Materials added in current process	70%	50,000	80%
Conversion costs	60%	<u>35,000</u>	40%
Total:		<u>185,000</u>	

- (2) Costs incurred during last month:

Materials transferred-in	200,000 units at HK\$380,000
Materials added	HK\$186,930
Conversion costs	HK\$112,860

- (3) Normal loss was 5% of the materials transferred-in and 11,000 units were actually disposed with nil value during last month. Normal loss was apportioned between completed units and closing WIP.

Required:

- (a) Calculate the number of equivalent units for materials transferred-in, materials added and conversion costs for last month. (7 marks)
- (b) Calculate the cost per equivalent unit for materials transferred-in, materials added and conversion costs. (4 marks)
- (c) Calculate the cost of units transferred to next process, abnormal loss/ gain and closing WIP for last month. (8 marks)
- (d) Describe joint products and by-products. (3 marks)

Question 2 (28 marks – approximately 50 minutes)

Pinky Products Limited ("PPL") manufactures two products: Premium and Deluxe. Details of the two products for next year are estimated as follows:

	<u>Premium</u>	<u>Deluxe</u>
Production units	100,000	200,000
Machine hours per unit	2	1.5
Direct materials per unit	HK\$20	HK\$10
Direct labour per unit	HK\$30	HK\$60

A traditional absorption costing is used at present. Production overheads are absorbed on a machine hour basis. The estimated production overheads for next year are HK\$6,400,000.

An executive director of PPL has challenged the appropriateness of using the current overhead absorption system and has recommended that the activity-based costing ("ABC") system would provide more relevant and accurate product cost information. Further analysis was provided by the management accountant to facilitate the use of ABC to absorb production overheads for next year.

	HK\$
Costs relating to machinery	2,000,000
Costs relating to set up	1,800,000
Costs relating to materials handling	1,400,000
Costs relating to inspection	<u>1,200,000</u>
Total	<u>6,400,000</u>

	<u>Premium</u>	<u>Deluxe</u>
Number of set up	410	190
Number of materials handling	920	1,080
Number of inspection	670	330

The selling price for each product is set at a uniform mark-up of 25%.

Required:

- (a) Calculate the selling price per unit for each product for next year using the traditional system of absorbing production overheads on a machine hour basis. (8 marks)
- (b) Applying activity-based costing, calculate the cost driver rate for each of the four production overheads and selling price per unit for each product for next year. (16 marks)
- (c) Contrast the selling price per unit for each product for next year calculated in Questions 2(a) and 2(b) above. (4 marks)

Question 3 (20 marks – approximately 36 minutes)

Sapphire Company Limited ("SCL") manufactures an organic health food product and has recently introduced a system of standard costing. The standard variable cost for the product is as follows:

	HK\$
Direct materials (10 kgs at HK\$20 per kg)	200
Direct labour (3 hours at HK\$80 per hour)	240
Variable manufacturing overheads (3 hours at HK\$50 per direct labour hour)	150
	<u>590</u>

59,000 kgs raw materials were purchased at a cost of HK\$1,298,000 and 5,600 units of the health food product were produced during last month. Other actual data for last month were as follows:

Raw materials used	54,000 kgs
Direct labour paid (18,000 hours)	HK\$1,350,000
Actual direct labour hours worked	17,800 hours
Variable manufacturing overheads incurred	HK\$860,000

Variable manufacturing overheads are absorbed based on actual direct labour hour worked. Budgeted normal capacity expressed in direct labour hours for last month is 18,300 hours. SCL decides to calculate the price variance of direct materials at the time of receipt.

Required:

- (a) Calculate the price and quantity variances for direct materials. (4 marks)
- (b) Calculate the rate, idle time and efficiency variances for direct labour. (6 marks)
- (c) Calculate the spending and efficiency variances for variable manufacturing overheads. (4 marks)
- (d) Justify the policy of SCL to calculate direct materials price variance at the time of receipt. (2 marks)
- (e) Calculate the labour production volume ratio and labour capacity ratio. (4 marks)

Question 4 (10 marks – approximately 18 minutes)

- (a) Conglomerates are increasingly adopting total quality management ("TQM") in their manufacturing and servicing activities. Explain the objective of using TQM and state the four categories of cost of quality.

(4 marks)

- (b) Dorothea Investment Limited ("DIL") has two geographical divisions which operate as profit centres. The data relating to two new proposed separate projects of the two divisions are provided.

	Project for Shanghai Division	Project for Guangdong Division
Capital required for investment	HK\$3,000,000	HK\$2,000,000
Return on investment ("ROI")	14%	16%
Residual income ("RI")	HK\$60,000	HK\$80,000

The firm's required rate of return ("RRR") is 12%. The ROI of both divisions before taking into account of the new proposed projects are 15%. DIL has ample capital for both new projects.

Required:

- (i) Analyse whether either or both projects should be taken based on the information provided when the investing decisions are made centrally.

(2 marks)

- (ii) Justify whether the investing decisions in Question 4(b)(i) above will be different if the divisions are treated as investment centres and the divisional manager has the power to make all investment decisions within the division.

(4 marks)

* * * END OF EXAMINATION PAPER * *